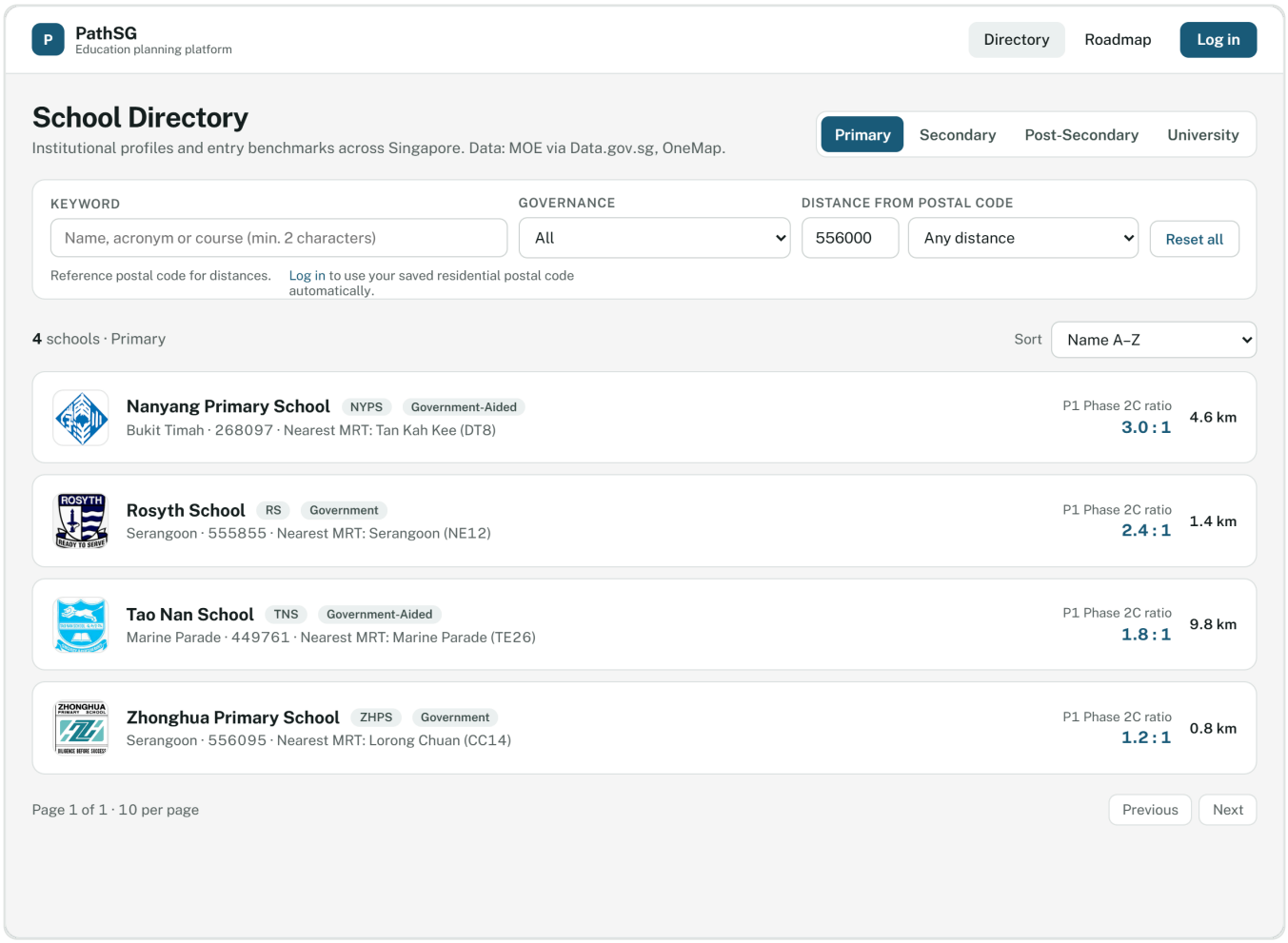
PathSG UI Mockup Walkthrough

Screen-by-screen documentation of the current PathSG interface mockup. Each figure is followed by a brief description of the screen and the use cases and requirements it addresses.

Five feature areas are covered: **1** School Directory & Profile Viewer, **2** User Account & Student Profile Management, **3** Education Roadmap Generator & Manager, **4** User School Review System, **5** Admin Manual Database Update. **UC** IDs refer to the Use Case Descriptions; **REQ** IDs to the functional requirements of the matching feature section, and PERF / SEC / QUAL / BR IDs to the non-functional requirements and business rules.

1 · School Directory

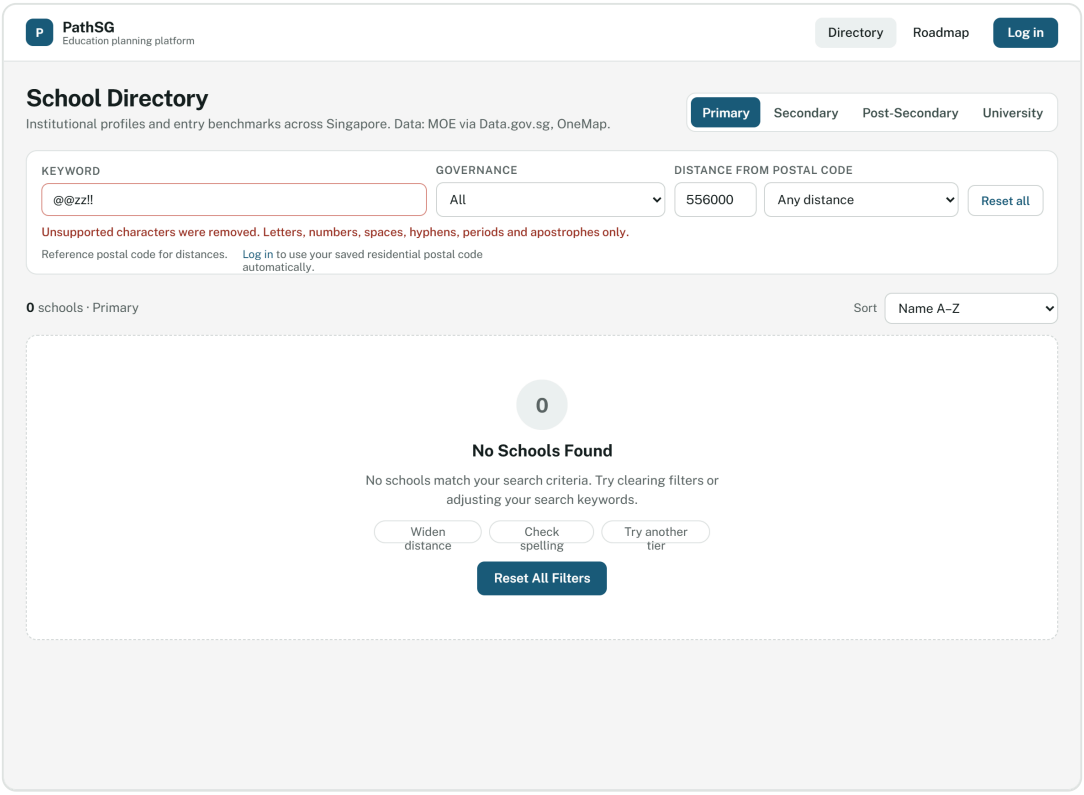
The landing view. A single-select tier switch (Primary / Secondary / Post-Secondary / University) drives a keyword search, governance and proximity filters, a sort control and a paginated result list at ten cards per page. Each summary card carries the school logo, official name, governance type, location, nearest MRT, the tier's primary entry benchmark and the distance from the reference postal code.



COVERS UC-1.1 · REQ-1.1 to 1.8 · PERF-4

2 · Zero-match empty state

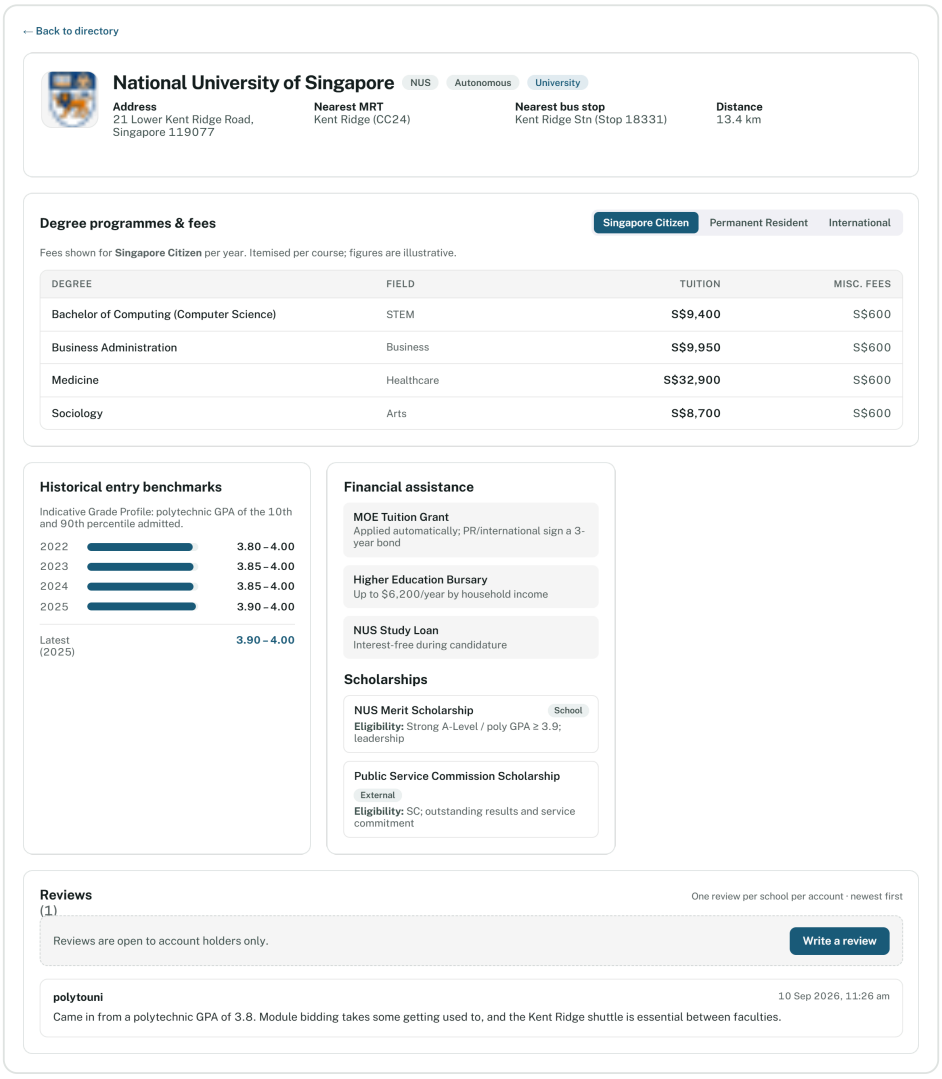
When a keyword or filter combination matches no institution the result list is replaced by an explicit “No Schools Found” state with the prescribed message, three suggestions for broadening the criteria and a single-click reset. Malformed input (special symbols, as typed here) is sanitised and falls through to the same state rather than erroring.



COVERS UC-1.1 AF-S4, AF-S5 · REQ-1.14, 1.15 · SEC-4

3 · School profile

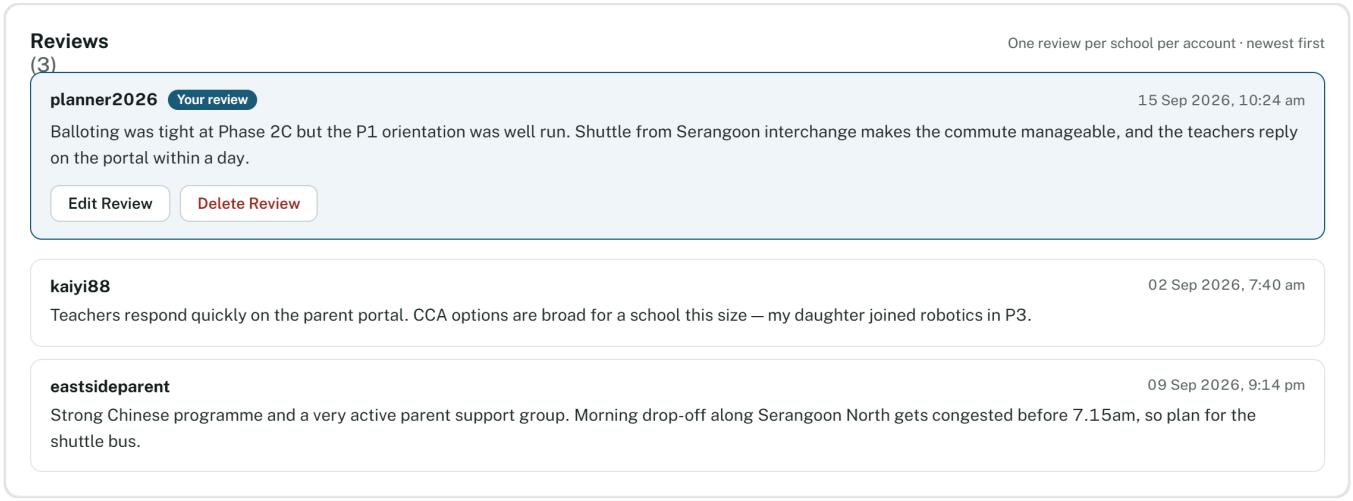
The full institutional record reached by selecting a summary card: identity and governance, address with postal code, nearest MRT and bus stop, the course list with itemised tuition and miscellaneous fees switchable across the three citizenship tiers, four years of entry benchmarks, and financial assistance and scholarships with stated eligibility. The reviews section documented next sits at the foot of this page.



COVERS UC-1.2 · REQ-1.9 to 1.13 · BR-1 · PERF-1

4 · School reviews

Reviews are read by everyone and written only by account holders, newest first, with the reviewer's username and submission timestamp. An account is limited to one review per school: once it exists, the composer is replaced by the author's own review card carrying Edit Review and Delete Review — deletion routes through a confirmation dialog, and editing reopens the same text area with a Cancel that discards changes. The composer counts words live and stops input at 200; blank or whitespace-only text is refused with the prescribed message, and HTML tags are stripped on submission. A school with no reviews shows a “No reviews yet” panel.



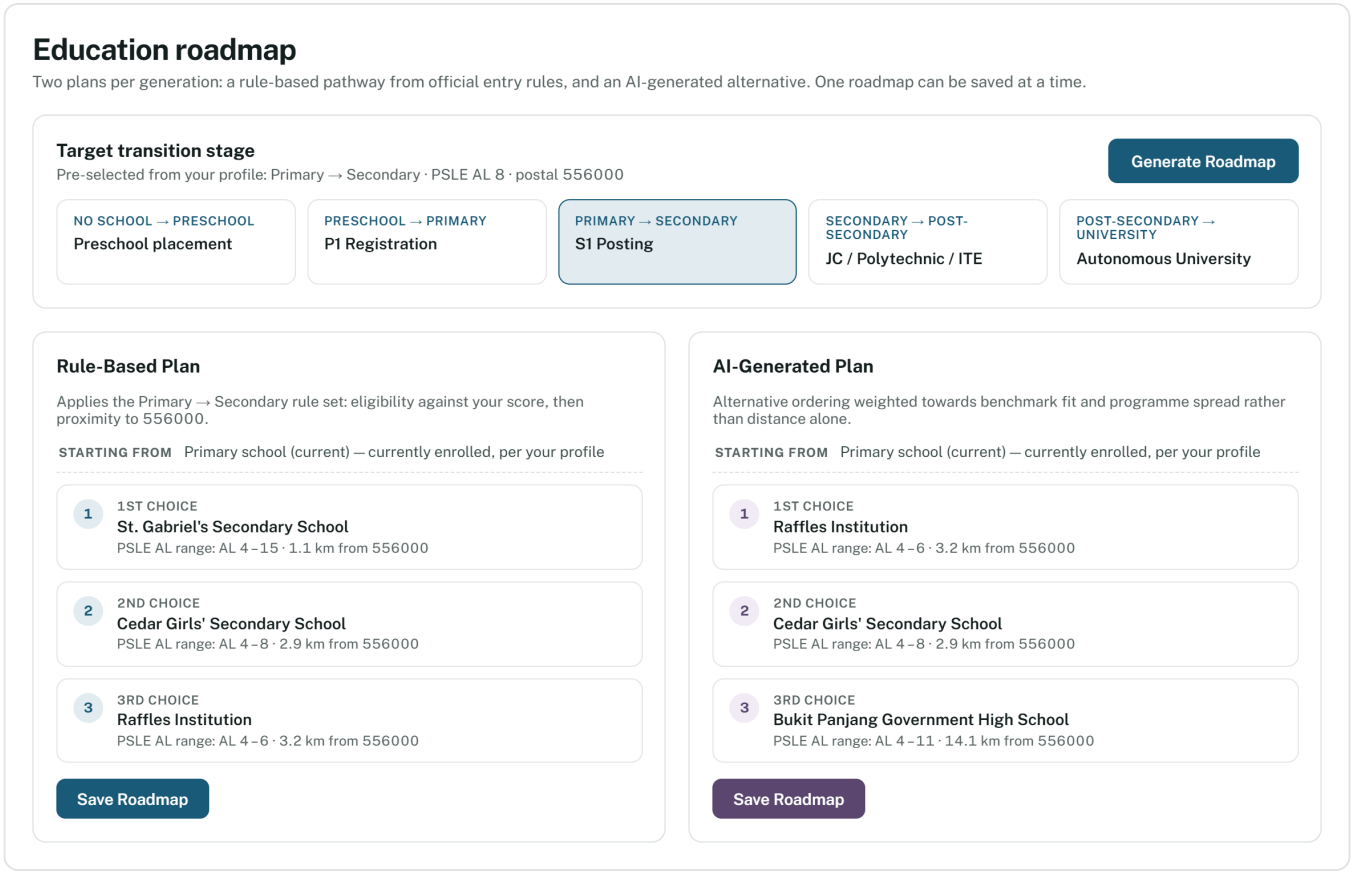
For a guest the composer is replaced by the gate below; Write a review raises the modal asking them to log in or create an account, leaving the existing reviews visible throughout.



COVERS UC-4.1 to UC-4.4 · REQ-4.1 to 4.11 · BR-6 · SEC-4 · QUAL-2

5 · Education roadmap — generation

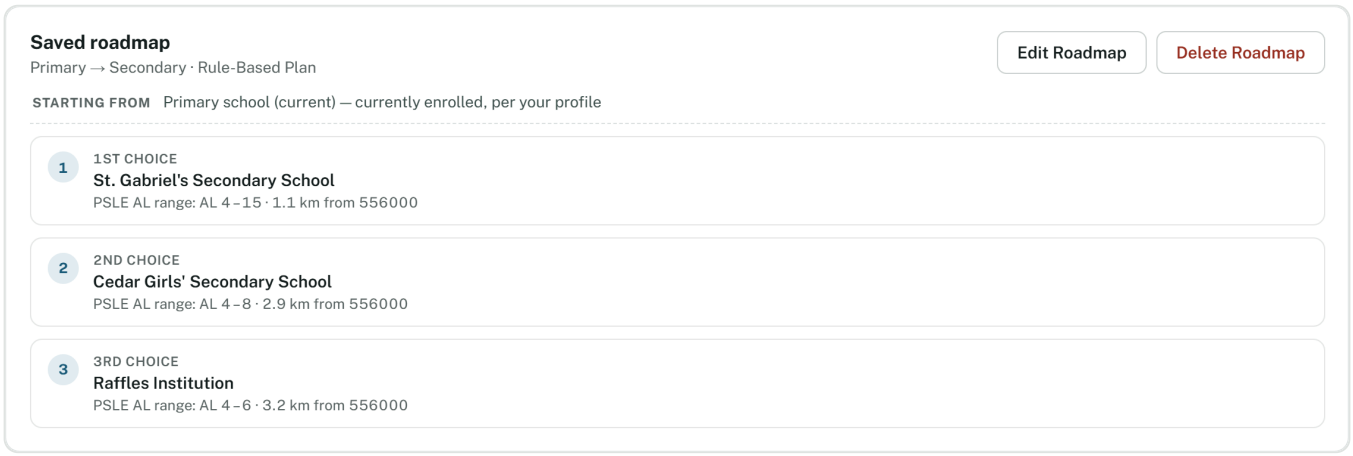
One generation produces two plans side by side. The target transition stage is pre-selected from the saved profile — the caption above the tiles states the stage, score and postal code it was drawn from — and any of the five stages can be chosen instead, each with its own entry rule set. The rule-based panel renders immediately; the AI panel runs asynchronously and shows its own loading state, so a failure there cannot delay or blank the rule-based plan. If the AI call errors or exceeds its threshold, the failure is confined to that panel with the prescribed message and a Retry AI Generation control. Generation is blocked before it starts if no valid postal code is stored, with a popup linking to the profile page.



COVERS UC-3.1 · REQ-3.1, 3.2, 3.3, 3.7, 3.8, 3.9, 3.12 · PERF-6, PERF-7 · QUAL-6, QUAL-7

6 · Education roadmap — saved plan

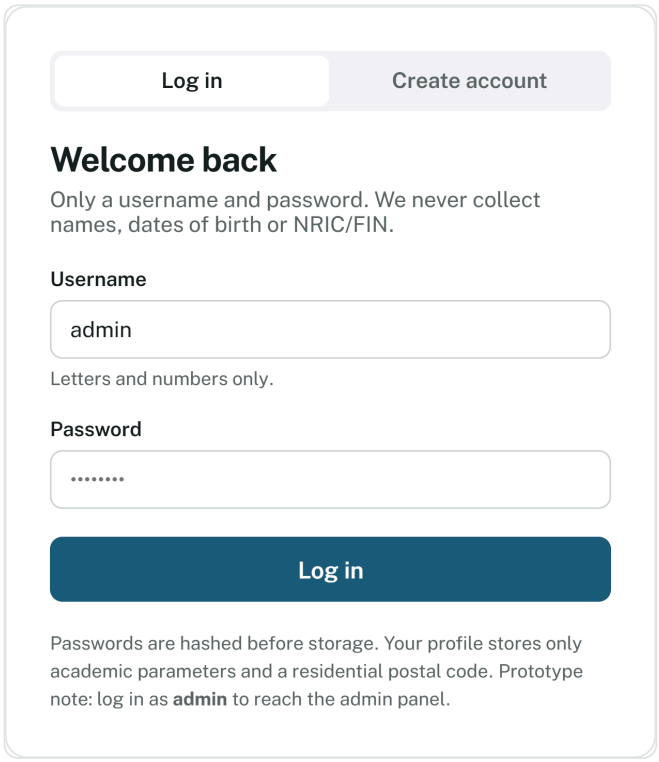
Saving either plan stores one roadmap against the account, labelled with its stage and the plan it came from; saving again while one exists asks for confirmation before overwriting. Returning to the page renders the saved roadmap with Generate, Edit and Delete controls. Edit Roadmap swaps each choice node for a dropdown restricted to eligible institutions, with Save Changes and a Cancel that restores the stored version; Delete Roadmap confirms first and returns the page to its empty state, which offers nothing but a Generate Roadmap button.



COVERS UC-3.2, UC-3.3, UC-3.4 · REQ-3.4, 3.5, 3.6, 3.10, 3.11, 3.13 to 3.16

7 · Log in

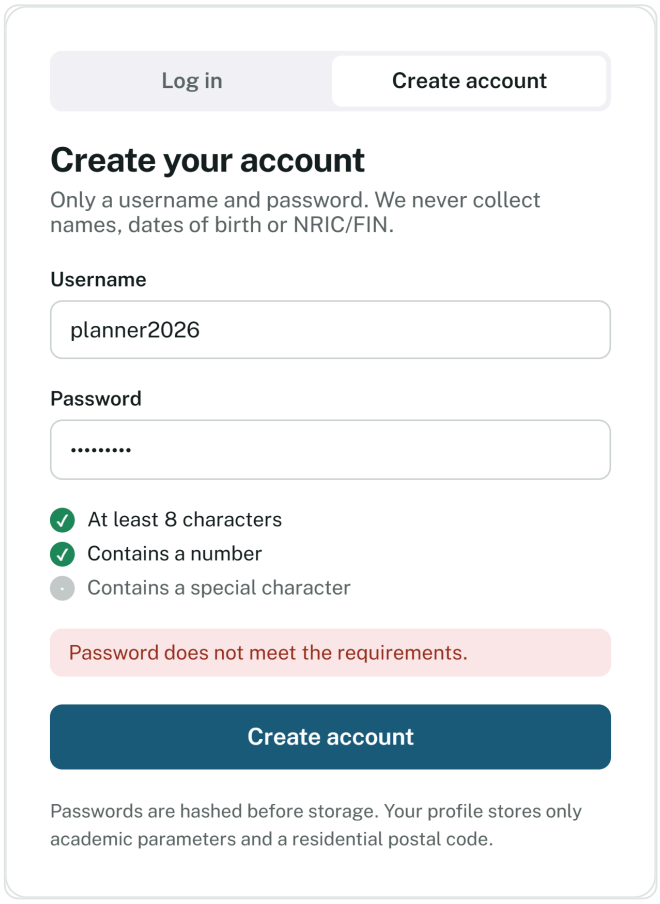
A guest who opens Roadmap, Saved or My profile is redirected to this card, with a banner above it stating why. Credentials are a username and password only, and the supporting copy states that nothing else is collected. Signing in as **admin** grants the Admin role and lands on the admin panel; any other account lands on the profile.



COVERS UC-2.2, UC-2.3 · REQ-2.1, 2.17 · BR-2 · SEC-5 · PERF-5

8 · Create account

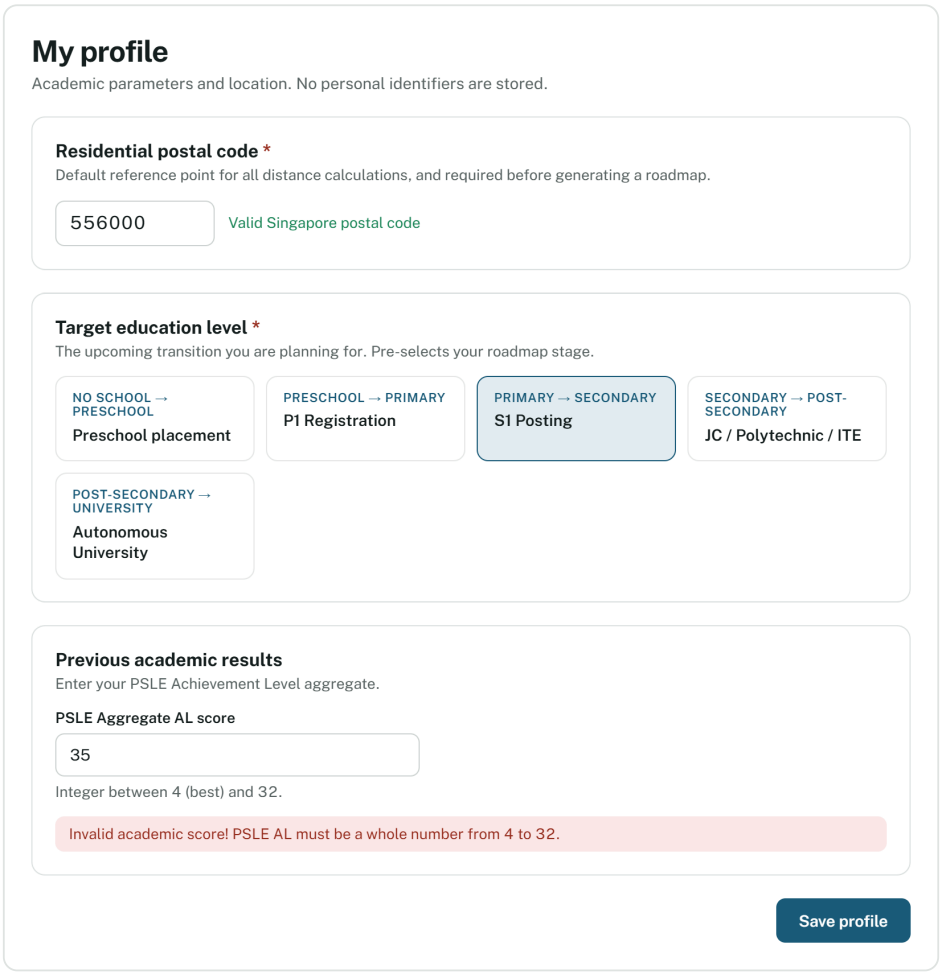
Registration shares the same card behind a tab. Password complexity is checked live against three rules (eight characters, one digit, one special character) and submission is blocked with an inline error until all three pass. Usernames already taken are refused. No name, date of birth or NRIC/FIN field exists in the form.



COVERS UC-2.1 · REQ-2.2 to 2.5, 2.15, 2.18 · SEC-1 to SEC-3 · QUAL-2

9 · Student profile

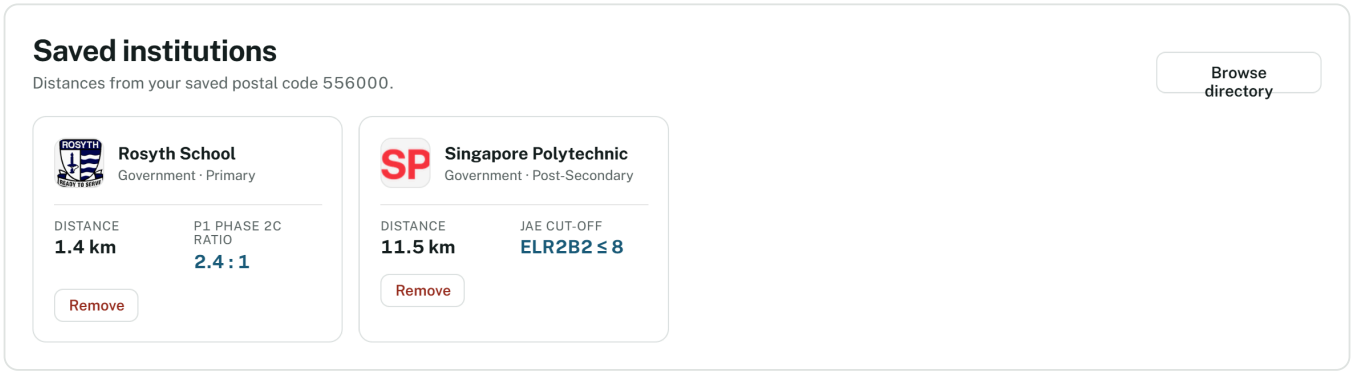
The residential postal code is validated as a six-digit string and becomes the anchor for every distance calculation in the system, and the precondition for roadmap generation. The target education level is stated as the upcoming transition, and it pre-selects the roadmap stage; choosing one re-renders only the academic fields relevant to that transition. Out-of-range scores block the save with an inline message, shown here for a PSLE AL aggregate of 35 against the permitted 4–32.



COVERS UC-2.4 · REQ-2.6 to 2.10, 2.16 · BR-3, BR-4 · PERF-2 · QUAL-2

10 · Saved institutions

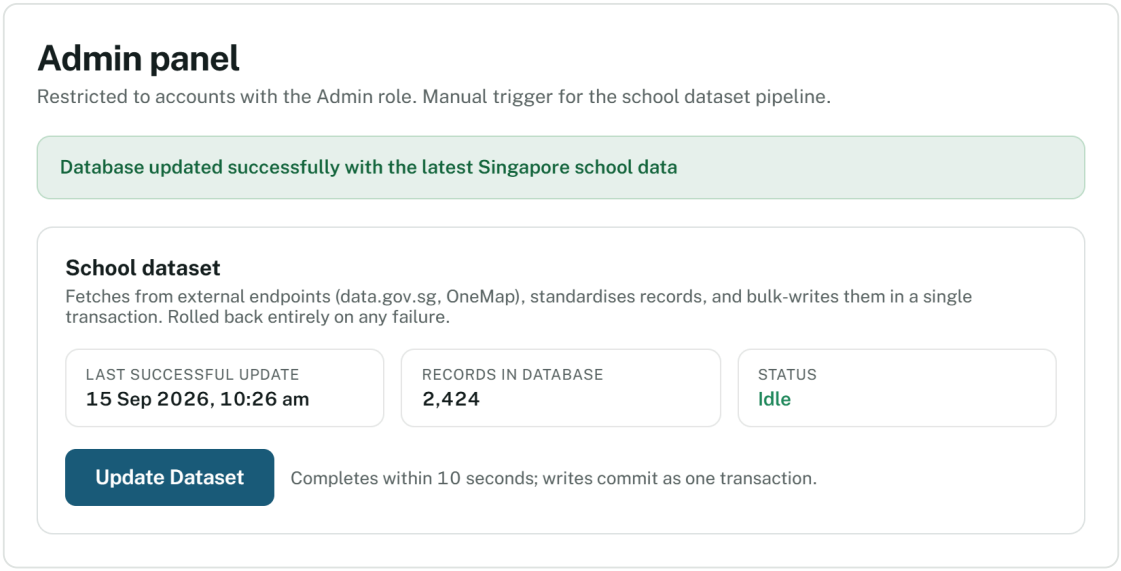
The bookmark dashboard, available only to an authenticated session. Each quick-view card shows the logo, official name, governance type and tier, the straight-line distance from the saved postal code and the tier's primary entry benchmark, with a single-click remove control. Bookmarks are added and removed from the star control on any summary card or profile.



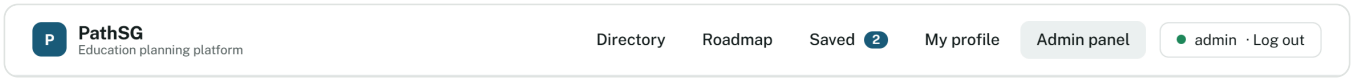
COVERS UC-2.5, UC-2.6, UC-2.7 · REQ-2.11 to 2.14 · BR-2 · PERF-3

11 · Admin panel

Reachable only by an account holding the Admin role, which lands on this screen directly at login. Update Dataset triggers the pipeline — fetch from the external endpoints, parse and standardise, then bulk-write in a single transaction — and disables itself for the duration to prevent a duplicate submission, showing the elapsed state in the button and the Status field. A commit prints the prescribed success banner and refreshes the last-update timestamp and record count above it. A timeout or a parsing failure instead aborts, leaves the existing records untouched, re-enables the button and raises the corresponding popup.



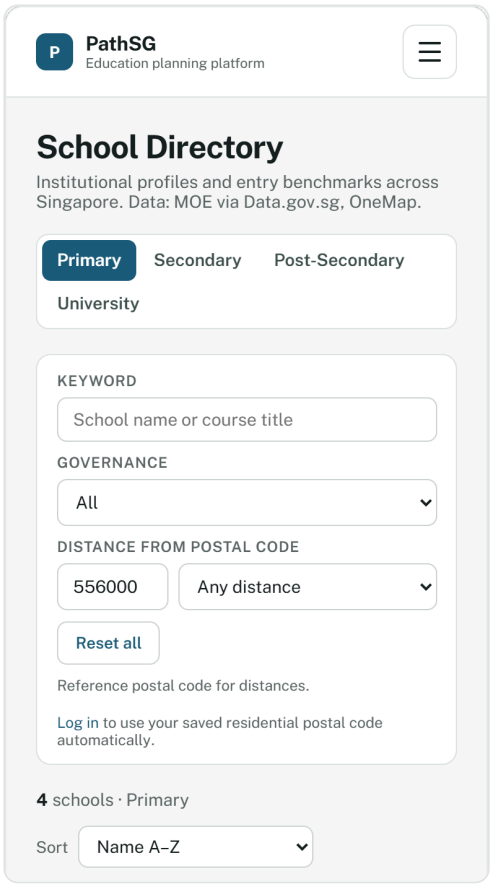
Access is enforced in the navigation as well: the Admin panel item appears only for an admin session, alongside the items any authenticated user sees.

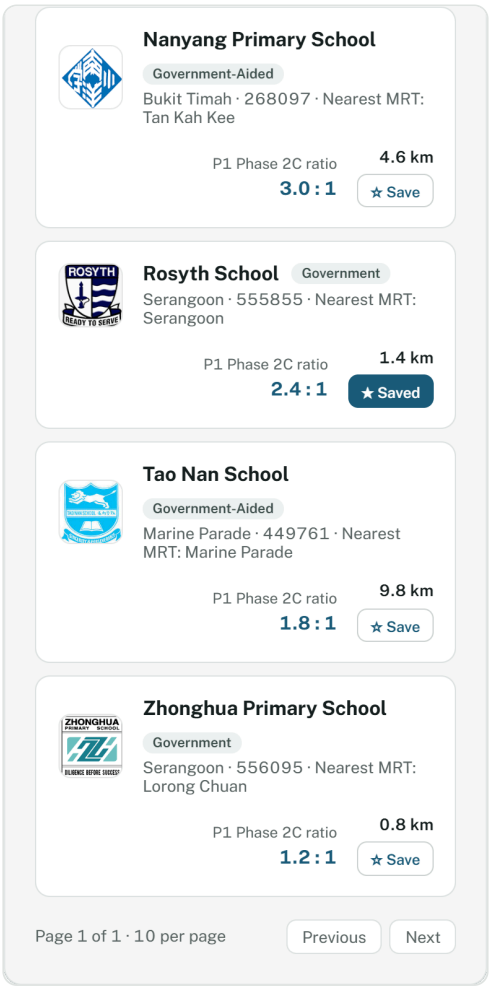


COVERS UC-5.1 · REQ-5.1 to 5.9 · BR-5 · PERF-8

12 · Mobile layout

The directory at 390 px, shown in two halves. Navigation collapses behind a menu button — including the role-gated items — the tier switch and filter panel stack to full width, and card content reflows to two lines, with no horizontal scrolling, no broken elements and touch targets held at 44 px.





COVERS QUAL-1